

Jesica Bermudes

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EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Computer Science

Expected Graduation: May 2027

- GPA: 3.4/4.0 | Dean's List (Spring 2026)
- Relevant Coursework: Data Structures & Algorithms I & II, Object-Oriented Programming, System Software, Operating Systems, Machine Learning, Cybersecurity, Discrete Mathematics, Linear Algebra, Statistics

TECHNICAL SKILLS

Languages: TypeScript, Python, C, Java, JavaScript

Frameworks & Libraries: React, Django, Node.js, Express.js, FastAPI, Tailwind CSS, pandas, HTML/CSS

Databases: PostgreSQL, SQL

Tools & Cloud: Git, GitHub, AWS, VS Code, PyCharm, Jupyter Notebook

Certifications: AWS Cloud Security Foundations, GitHub Foundations

PROJECTS

Hospital Resource Allocation Simulator | *TypeScript, React, Node.js, Express.js*

- Built a **full-stack discrete-event simulator** modeling Emergency Severity Index (ESI) triage, patient arrivals, treatment flow, and contention for limited staff and equipment across normal, flu-surge, and mass-casualty scenarios.
- Designed an **all-or-nothing multi-resource allocator** that verifies staff and equipment availability before treatment and reassesses waiting patients as resources become available.
- Implemented first-come-first-served (FCFS), priority, and priority-aging schedulers with **seeded patient generation** for reproducible comparisons.
- Evaluated schedulers across five 10,000-patient runs; priority aging achieved a **~88% reduction in maximum wait time** versus priority-only scheduling.

BubbleFlow | *Python, Django REST Framework, PostgreSQL, React, TypeScript*

- Built a **full-stack prepared-inventory and production system** based on firsthand experience as a boba shop shift lead, linking front-of-house orders to back-of-house inventory and production needs.
- Designed an **append-only inventory ledger** in PostgreSQL that derives batch balances from production, consumption, waste, correction, and expiration events, preserving a full audit trail.
- Implemented **FEFO batch consumption with transactional row locking** to use earlier-expiring stock first and prevent overselling during concurrent inventory operations.
- Developed a **demand-aware production recommendation engine** using recent consumption rates, current stock, incoming batches, and configurable safety buffers.

HTTP/1.1 & WebSocket Server | *C, POSIX Sockets, TCP, Linux epoll*

- Built an **event-driven HTTP/1.1 and WebSocket server in C** using non-blocking POSIX sockets and Linux epoll, without HTTP or networking frameworks.
- Implemented **incremental HTTP request parsing**, persistent connections, static file serving, status-code handling, and bounded per-connection input/output buffers.
- Designed a **single-threaded connection state machine** to handle partial reads and writes across concurrent clients without blocking the event loop.
- Implemented the **WebSocket upgrade and framing protocol**, including masked client frames, ping/pong, connection closing, and real-time message broadcasting.

EXPERIENCE

Jade Tea House

Davie, FL

Shift Lead

Aug. 2022 – Dec. 2024

- Led shifts of up to five employees in a high-volume beverage environment, delegating tasks and coordinating order flow during peak demand exceeding 300 orders per day.
- Planned daily prep by determining stock needs, monitoring inventory levels, waste, and expiration of prepared items to keep the line supplied.
- Cross-trained across all stations and onboarded new hires on workflow procedures, communication expectations, and drink preparation standards.
- Contributed to menu development by testing new recipes and incorporating customer feedback into product improvements.